

SANCTI SPIRITUS, Cuba

WMO: 783490

Lat: 21.933N

Lon: 79.450W

Elev: 318

StdP: 14.53

Time Zone: -5.00 (CUB)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.2	58.3	48.7	51.5	62.3	51.5	57.1	64.9	20.7	72.7	18.6	74.0	2.4	320	0.245

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.7	91.6	76.8	90.4	76.5	89.4	76.3	79.7	87.1	79.1	86.7	78.5	86.1	4.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	144.6	83.2	77.0	142.2	82.7	76.4	139.5	82.1	43.4	87.5	42.7	87.0	42.2	86.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	13.9	11.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.3	72.3	73.5	73.9	77.6	79.1	80.5	81.4	81.7	80.5	78.4	75.6	72.7
	DBStd	4.52	4.07	4.07	4.15	2.72	2.13	2.08	1.72	1.64	1.55	2.38	3.28	4.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	16	5	3	2	0	0	0	0	0	0	0	0	6
	CDD50	9960	692	659	741	829	901	914	973	984	915	881	768	703
	CDD65	4501	232	242	278	379	436	464	508	519	465	416	318	244
	CDH74	34398	1297	1580	1997	2994	3421	3902	4584	4780	3735	2915	1935	1258
	CDH80	11382	260	404	572	1054	1232	1366	1783	1926	1282	834	447	222
(14) Wind	WSAvg	3.7	4.3	4.3	4.8	4.4	3.6	2.9	3.0	3.0	2.7	3.2	3.8	4.0
(15) Precipitation	PrecAvg	58.7	1.3	1.9	1.8	2.2	9.3	10.1	6.3	7.9	8.3	7.3	2.1	1.0
	PrecMax	78.1	4.8	7.4	9.4	9.0	24.5	19.8	12.1	26.1	13.2	21.3	6.8	4.6
	PrecMin	42.6	0.0	0.0	0.1	0.0	2.8	3.2	1.9	0.0	3.3	1.5	0.0	0.0
	PrecStd	9.1	1.2	2.0	1.7	1.9	5.6	4.4	2.8	4.3	2.5	4.4	1.8	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	87.9	89.6	91.1	91.8	91.6	92.9	93.8	90.8	89.4	87.6	86.2
		MCWB	71.5	70.9	71.4	72.0	74.1	78.0	77.7	77.2	77.2	77.6	75.2	72.6
	2%	DB	84.6	86.2	87.3	89.0	89.9	89.9	91.1	91.6	89.4	87.8	85.8	84.3
		MCWB	71.1	70.6	71.0	71.9	73.5	77.6	77.4	77.7	77.6	77.2	75.2	72.5
	5%	DB	83.1	84.6	85.5	87.7	88.4	88.5	89.8	90.2	88.2	86.4	84.4	82.5
		MCWB	71.0	70.5	70.9	72.1	73.3	77.2	77.5	77.6	77.6	76.9	74.8	72.0
	10%	DB	81.1	82.7	83.5	86.0	86.6	87.0	88.2	88.8	86.9	84.9	82.7	80.6
		MCWB	70.4	70.4	70.5	71.6	73.4	76.8	77.4	77.5	77.6	76.6	74.2	71.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.5	74.2	75.9	77.3	78.2	79.9	80.1	80.6	80.1	79.4	78.0	75.4
		MCDB	81.7	81.7	82.5	83.2	87.1	87.2	87.9	88.2	86.5	85.7	84.4	81.3
	2%	WB	73.3	72.9	74.2	75.3	77.0	79.0	79.1	79.6	79.2	78.5	76.7	74.4
		MCDB	80.5	80.7	81.9	82.2	84.3	86.6	87.3	87.4	86.2	85.0	82.7	80.3
	5%	WB	72.5	72.4	72.9	74.3	76.0	78.2	78.5	78.9	78.6	77.9	76.0	73.5
		MCDB	79.1	80.0	80.4	81.9	83.1	85.6	86.6	86.9	85.8	84.2	82.0	79.5
	10%	WB	71.6	71.6	72.1	73.4	75.1	77.6	77.9	78.2	77.9	77.2	75.1	72.6
		MCDB	77.8	79.1	79.8	81.7	82.0	84.6	85.7	86.0	84.8	83.3	81.0	78.2
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	18.5	18.6	17.7	15.8	13.9	15.3	15.7	14.4	13.3	14.3	15.6
		MCDBR	18.5	19.7	20.3	19.1	18.5	15.5	16.8	17.3	15.7	15.3	15.7	16.7
		MCWBR	6.7	6.4	6.4	5.1	5.2	5.5	5.7	5.9	5.7	6.1	6.4	6.5
	5% WB	MCDBR	16.5	17.7	17.0	16.0	14.9	14.2	15.5	15.9	14.8	13.7	13.9	15.3
		MCWBR	6.9	6.6	6.6	5.6	5.5	5.8	6.2	6.4	6.2	6.2	6.4	6.7
(40) Clear-Sky Solar Irradiance	taub		0.359	0.366	0.382	0.403	0.425	0.459	0.471	0.442	0.417	0.402	0.381	0.377
	taud		2.523	2.503	2.446	2.381	2.367	2.300	2.246	2.354	2.433	2.466	2.477	2.488
	Ebn at Noon		289	294	292	286	277	265	262	271	278	279	279	279
	Edh at Noon		30	33	36	39	40	42	44	40	36	34	32	31
(44) All-Sky Solar Radiation	RadAvg		1336	1620	1881	2060	1982	1900	2064	1988	1777	1528	1369	1248
	RadStd		72	61	70	61	189	127	108	70	91	108	80	91

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page